

- 9 Which row shows the correct formula for the named substance?

|          | substance           | formula                  |
|----------|---------------------|--------------------------|
| <b>A</b> | cobalt(II) chloride | $\text{Cu}_2\text{Cl}$   |
| <b>B</b> | sodium carbonate    | $\text{Na}_2\text{CO}_3$ |
| <b>C</b> | xenon               | $\text{Xe}_2$            |
| <b>D</b> | ammonium sulfate    | $\text{NH}_4\text{SO}_4$ |

- 10 The equation shows the thermal decomposition of magnesium carbonate.



[ $M_r$ :  $\text{MgCO}_3$ , 84]

Which mass of magnesium oxide is formed when 21.0 g of magnesium carbonate is completely decomposed?

- A** 1.9 g                      **B** 4.0 g                      **C** 10.0 g                      **D** 40.0 g
- 11 Concentrated aqueous sodium chloride is electrolysed using inert electrodes.

What is the main product formed at the positive electrode (anode)?

- A** chlorine  
**B** hydrogen  
**C** oxygen  
**D** sodium

- 12 A hydrogen–oxygen fuel cell uses  $630 \text{ dm}^3$  of oxygen.

The oxygen for the reaction is extracted from clean, dry air.

What is the minimum volume of clean, dry air needed to provide this volume of oxygen?

- A**  $788 \text{ dm}^3$                       **B**  $808 \text{ dm}^3$                       **C**  $3000 \text{ dm}^3$                       **D**  $3316 \text{ dm}^3$